

Lab sheet 03**Submission Due: 18.06.2024****Title:** Introduction to Database and MySQL (OOP).**Aims:**

- Setting up a MySQL database
- Writing SQL queries for CRUD operations
- Basic Page

Tasks:

- Create database in MySQL
- Build the connection with PHP
- Create / Store data in MySQL database using PHP
- Read data in MySQL database using PHP
- Update in MySQL database using PHP
- Delete in MySQL database using PHP

Activities :

1. Create database in MySQL
2. Implement connection

```
private function connect()
{
    $this->connection = mysqli_connect($this->host, $this->
username, $this->password, $this->database);
    if (!$this->connection) {
        die("Connection Failed: " . mysqli_connect_error());
    }
}
```

3. Read Operation.

```
public function getAllStudents()
{
    $query = "select * from `students`";
    $result = mysqli_query($this->connection, $query);

    if (!$result) {
        die("Failed to retrieve data");
    }

    $students = [];
    while ($row = mysqli_fetch_assoc($result)) {
        $students[] = $row;
    }
}
```

```
    }  
  
    return $students;  
}
```

4. Create / Store / Insert Operation

```
public function create()  
{  
    if (empty($this->name) || empty($this->gpa)) {  
        return header('location:index.php?error=All fields are  
required');  
    }  
  
    $query = "insert into `students` (`name`,`gpa`) values  
('$this->name', '$this->gpa')";  
    $result = mysqli_query($this->connection, $query);  
  
    if (!$result) {  
        return header('location:index.php?error=Failed to add  
student');  
    }  
  
    return header('location:index.php?success=Successfully added  
student');  
}
```

5. Update Operation

I. Read specific student

```
public function read()  
{  
    $query = "select * from `students` where `id` = '$this->  
>id'";  
    $result = mysqli_query($this->connection, $query);  
  
    if (!$result) {  
        return false;  
    }  
    $row = mysqli_fetch_assoc($result);  
    $this->setId($row['id']);  
    $this->setName($row['name']);  
    $this->setGpa($row['gpa']);  
    return true;  
}
```

II. Update the student data

```
public function update()
{
    if (empty($this->id) || empty($this->name) || empty($this->gpa)) {
        return header('location:index.php?error=All Fields are required');
    }

    $query = "update `students` set `name` = '$this->name',
`gpa` = '$this->gpa' where `id` = '$this->id'";
    $result = mysqli_query($this->connection, $query);

    if (!$result) {
        return header('location:index.php?error=Failed updated student data');
    }

    return header('location:index.php?success=Successfully updated student data');
}
```

6. Delete operation

```
public function delete()
{
    $query = "delete from `students` where `id` = '$this->id'";
    $result = mysqli_query($this->connection, $query);

    if (!$result) {
        return header('location:index.php?error=Failed to delete student');
    }

    return header('location:index.php?success=Successfully deleted student');
}
```

7. Add some CSS to the web page using [Bootstrap](#).

Exercise:

1. Create new database name as “todo”
2. Create new table “task”
3. That table contain id (primary key) , title and description.
4. Design a sample web page for this ToDo application using html and bootstrap.
5. Perform CRUD operation.

Discussion:

- MySQLi extension
- Limit data selection from s MySQL database
- Multiple record insertion to MySQLdatabase